

Trae IDE Agent Skills Case Study

Automating Onboarding & Management Workflows

Project Overview

- **Objective**

- Streamline new employee onboarding, organizational management, and team enablement using AI Agents.

- **Key Technologies**

- Trae IDE, Claude Agent Skills, Python (Pandas), HTML/JS/Tailwind.

Skill 1: Intelligent Excel Processing

- **Skill Used: `xlsx` / `RunCommand`**
- Challenge: Managing complex organizational charts and training data.
- Actions Taken:
 - Analyzed "中欧总表.xlsx" to extract trainee progress (Zou Fengjing: 4%).
 - Batch replaced sensitive personnel names with "Pending" (待定) using Regex.
 - Converted tabular data into Mermaid structure diagrams.
 - Outcome: Automated data hygiene and visualization preparation.

Skill 2: Workflow Automation

- **Skill Used: outlook-sender**
- Challenge: Manual email notifications for reports.
- Actions Taken:
 - Created a custom skill to interface with local Outlook via PowerShell.
 - Designed scripts for sending emails with attachments and saving drafts.
 - Outcome: Standardized communication capability (demonstrated proof-of-concept).

Skill 3: Frontend Data Visualization

- **Skill Used: frontend-design**
- Challenge: Text-based status reports are hard to digest.
- Actions Taken:
 - Transformed Markdown status reports into an interactive HTML dashboard.
 - Implemented real-time progress charts using Chart.js.
 - Added interactive features: Role Detection & Weight Adjustment.
 - Outcome: Professional, responsive onboarding status board.

Skill 4: Knowledge Enablement

- **Skill Used: enjoy-bot-community-guide**
- Challenge: Team lacks best practices for using AI community resources.
- Actions Taken:
 - Encapsulated community guidelines into a retrievable Skill.
 - Defined "A-B-C" usage strategy (Scenarios, Structure, Action).
 - Standardized "High-Quality Question Template".
 - Outcome: Scalable knowledge transfer mechanism.

Conclusion

- **Impact of Agent Skills:**
 - Structured: Converting ad-hoc tasks into reusable modules.
 - Automated: Reducing manual effort in data processing and reporting.
 - Interactive: Moving from static text to dynamic visual tools.
- This project demonstrates how Trae IDE Agents can act as full-stack engineers to handle end-to-end workflows.